Truck Fleet Failure Prediction

Team BrakeGuard

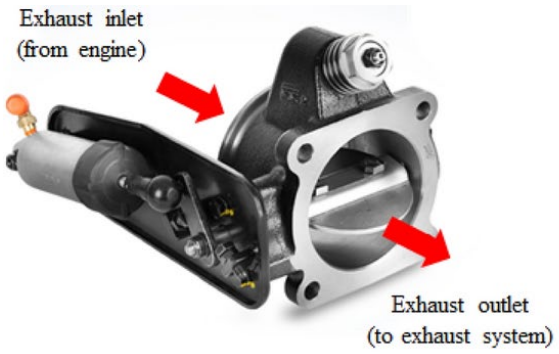
guga-fayrooz-seungdo-vishred

Exhaust Brake

Exhaust Brake Butterfly Valve

Field Observation

- The vehicles in the field were showing a safety problem.
- When the driver tries to put the brake, some vehicles do not brake properly.
- The exhaust brake reduces the speed of the vehicle. It can be used as a supplementary braking system, which reduces the effort and increases the life of the service brakes by 2 to 3 times.
- However, this component valve is sticking.
- We have verification of what valves were sticking as of May 2021.

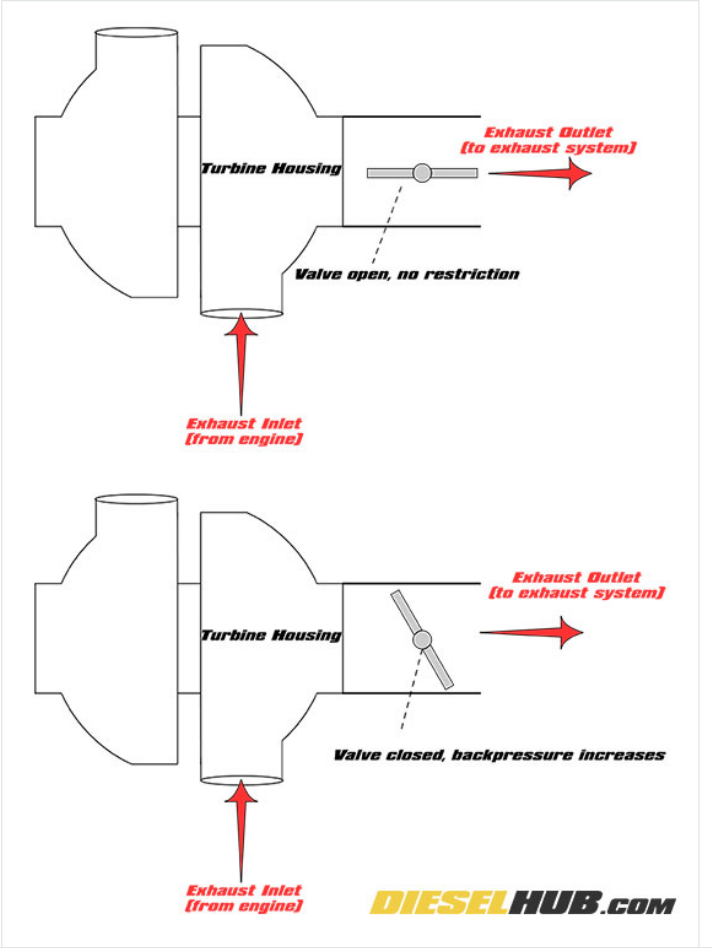


(a)

(b)

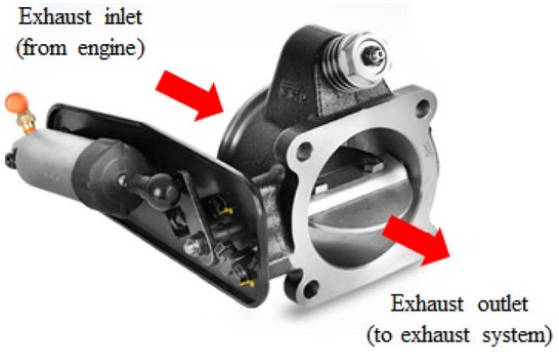
UNRESTRICTED

Exhaust Brake Function



No exhaust Brake

During the Braking, this valve should be closed.
If this is **stuck open**, The exhaust brake does not work.

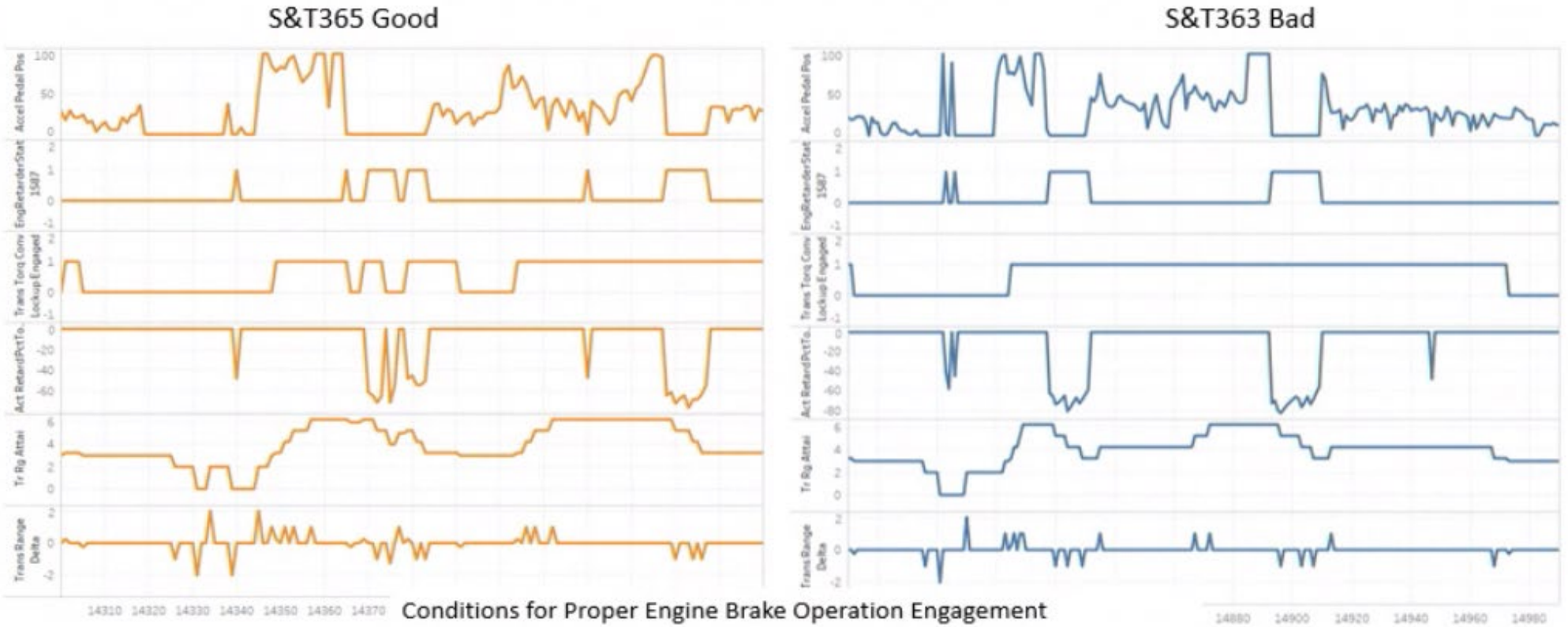


(a)

(b)

Exhaust Brake start to work

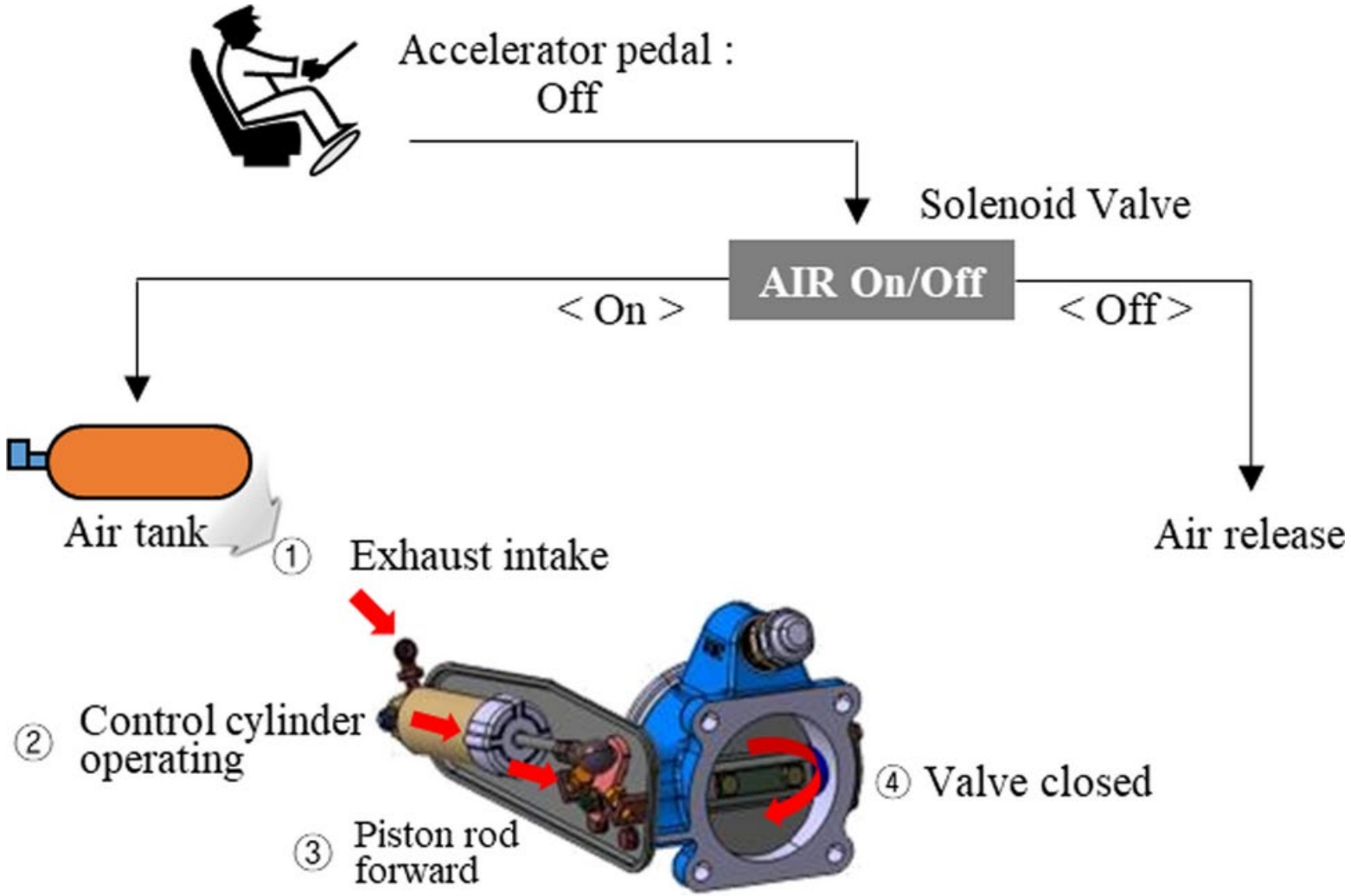
Field Technician's Feedback on Good vs Bad Brake



Conditions for Proper Engine Brake Operation Engagement

1. Trans converter lockup engaged=1 for at least 2 seconds, and
2. Accel Pedal Position<1, and
3. Act Retard PCT Torque Exhaust<0 for at least 1 second, and
4. Engine Retarder Stat 1587=1

Braking Method



Select the right operating condition for Braking

Use **event samples** (not raw time series) and filter to rows where:

- Driver intent
  - AccelPedalPos ≤ 2% (or _1587)
  - EngPctTorq ≤ 5% (or EngDmdPctTorq ≈ 0)
- Powertrain state
  - EngRetarderStat_1587 == 1
  - TransTorqConvLockupEngaged == 1
- Kinematics
  - EngSpeed ∈ [1400, 2200] RPM
  - VehSpeedEng ∈ [20, 60] mph
  - TrRgAttai stable (no gear change in window)
- Exclusions
  - BrakeSwitch == 0 (optional)
  - ABS flags OFF (ABS_RetCont_1587 == 0, etc.)

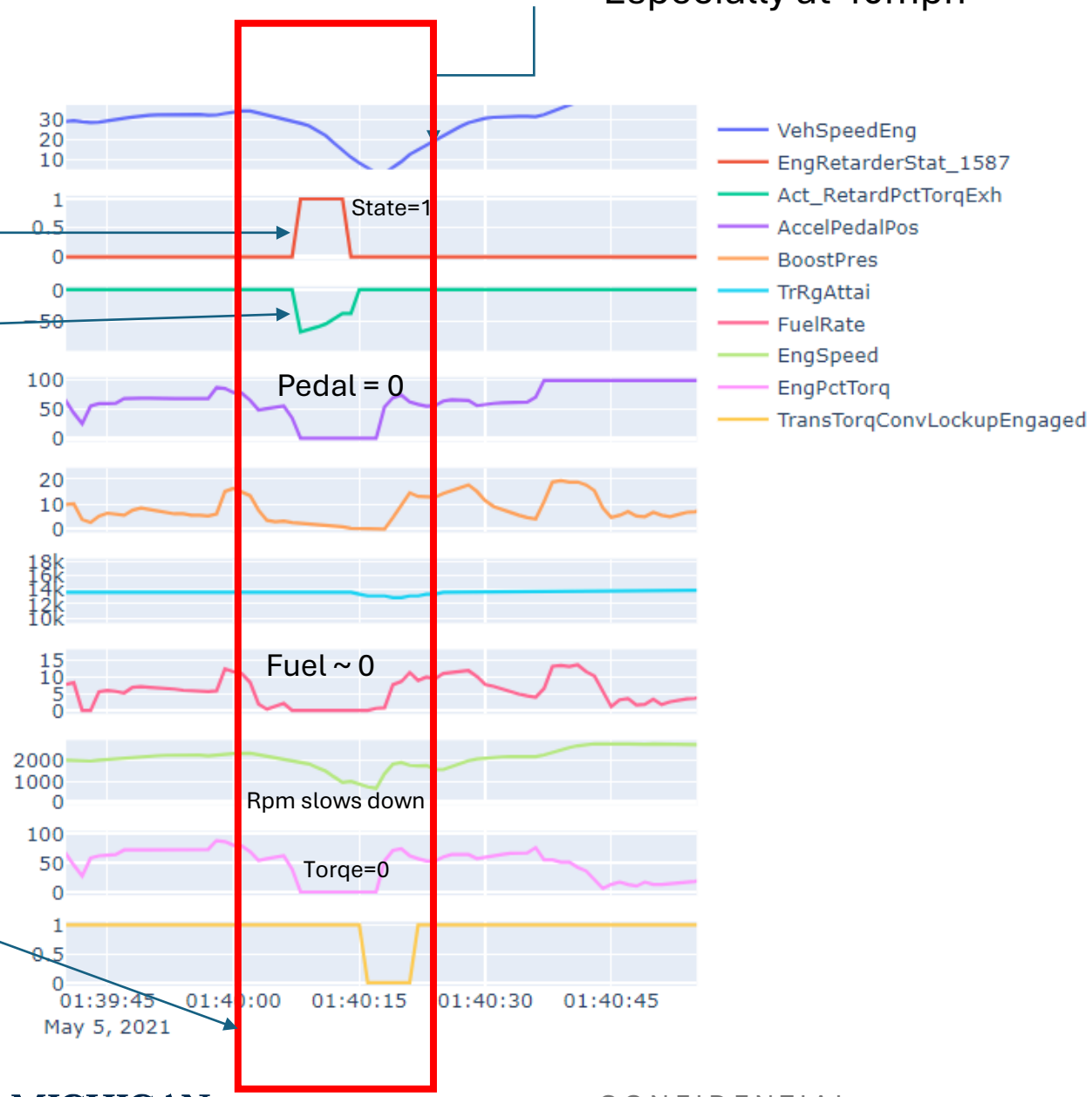
This isolates “engine-driven exhaust braking only”

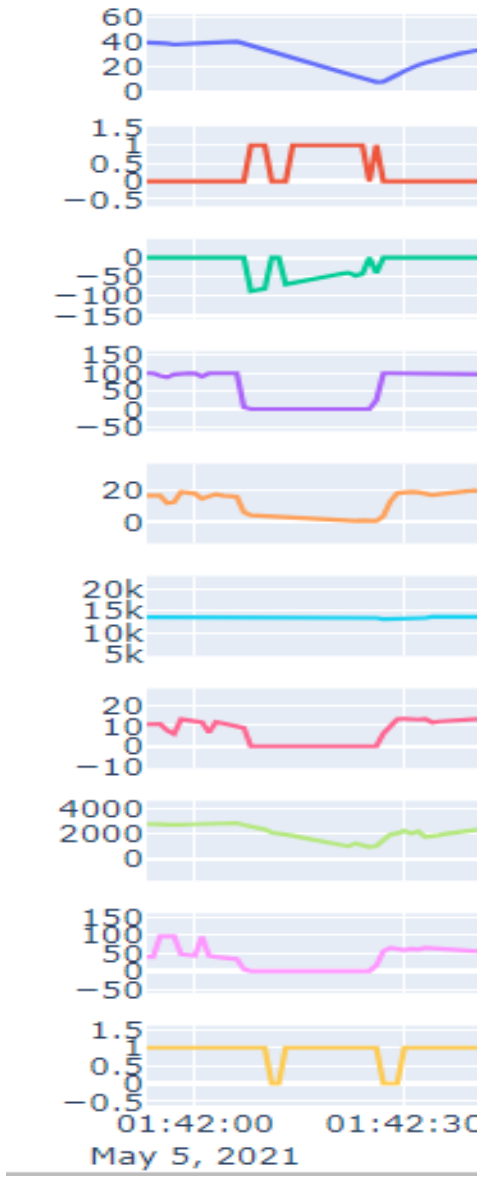
Signals

During the failure of the valve, the change in speed is slower
Especially at 40mph

Engine
Retarder is on

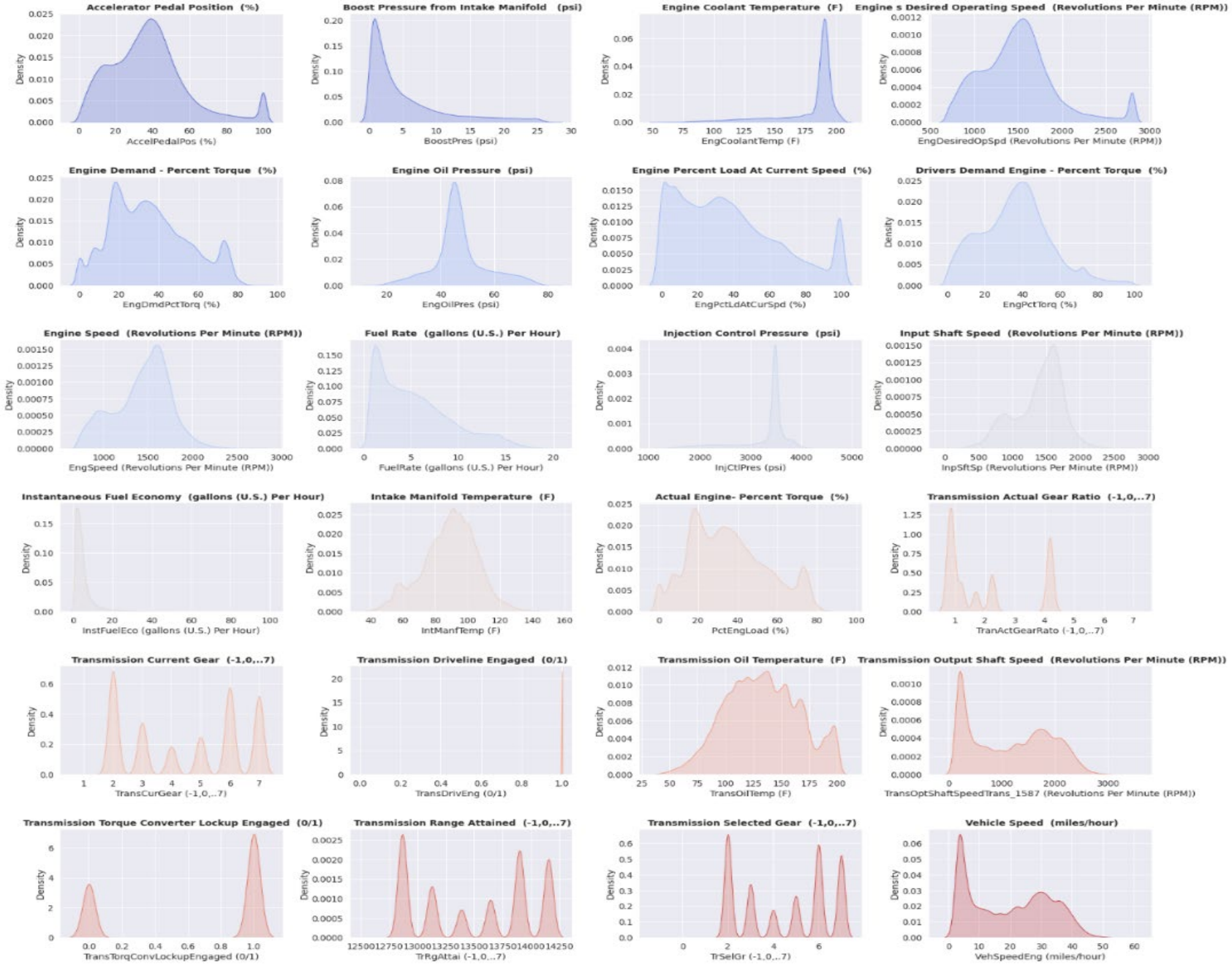
Exhaust back pressure
Is budling up and then
reduces due to zero fuel





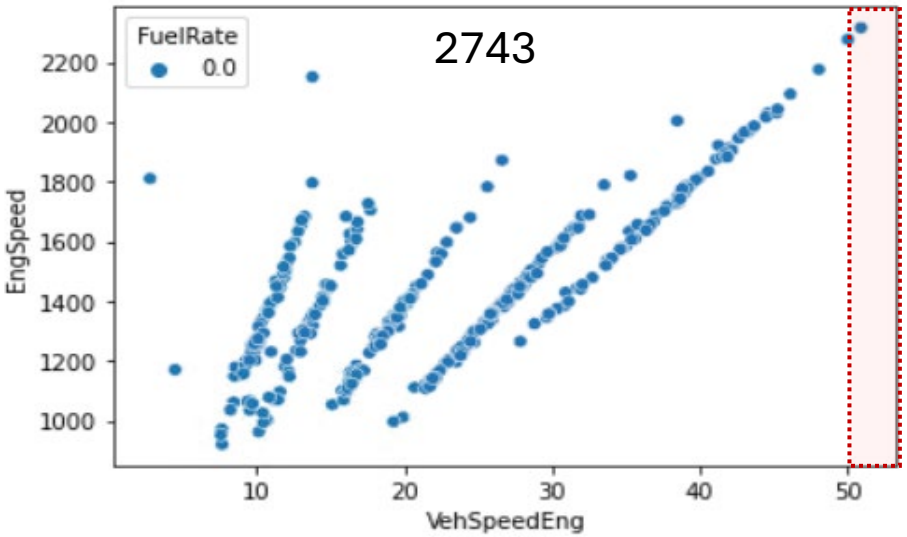
Details in this window of data

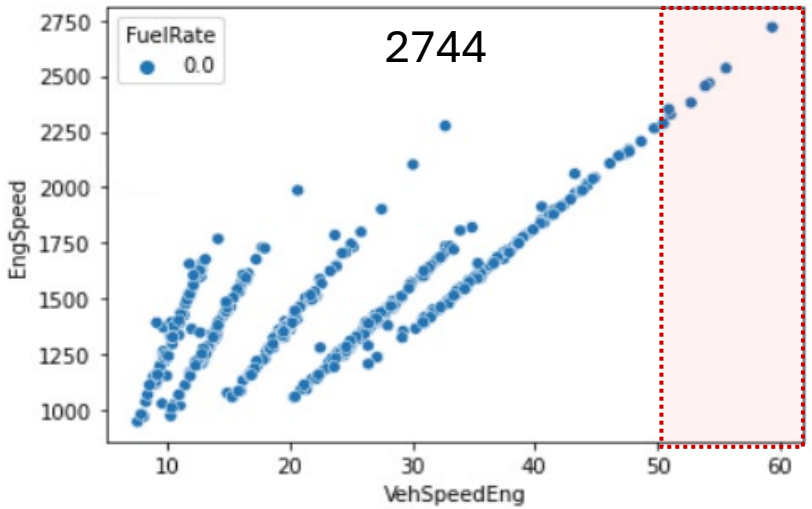
EDA

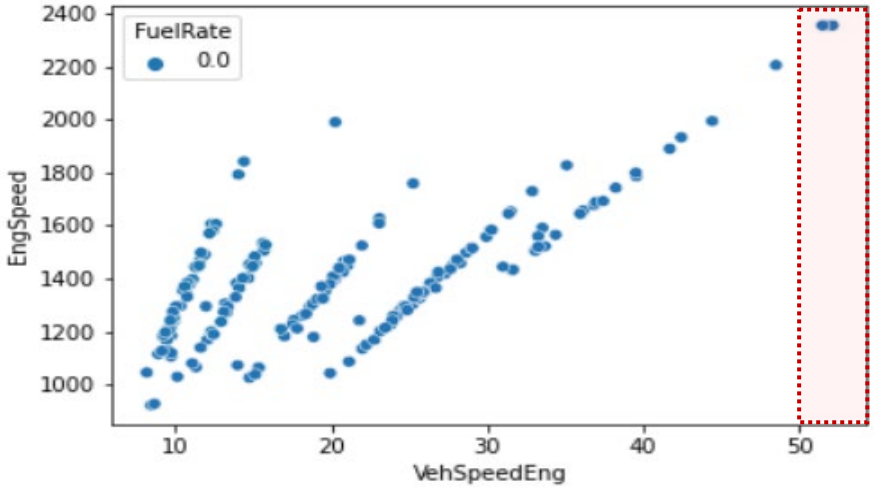

Accelerator Pedal Position (%)
AccelPedalPos (%)
Boost Pressure from Intake Manifold (psi)
BoostPres (psi)
Engine Coolant Temperature (F)
EngCoolantTemp (F)
Engine s Desired Operating Speed (Revolutions Per Minute (RPM))
EngDesiredOpSpd (Revolutions Per Minute (RPM))
Engine Demand - Percent Torque (%)
EngDmdPctTorq (%)
Engine Oil Pressure (psi)
EngOilPres (psi)
Engine Percent Load At Current Speed (%)
EngPctLdAtCurSpd (%)
Drivers Demand Engine - Percent Torque (%)
EngPctTorq (%)
Engine Speed (Revolutions Per Minute (RPM))
EngSpeed (Revolutions Per Minute (RPM))
Fuel Rate (gallons (U.S.) Per Hour)
FuelRate (gallons (U.S.) Per Hour)
Injection Control Pressure (psi)
InjCtlPres (psi)
Input Shaft Speed (Revolutions Per Minute (RPM))
InpSftSp (Revolutions Per Minute (RPM))
Instantaneous Fuel Economy (gallons (U.S.) Per Hour)
InstFuelEco (gallons (U.S.) Per Hour)
Intake Manifold Temperature (F)
IntManfTemp (F)
Actual Engine- Percent Torque (%)
PctEngLoad (%)
Transmission Actual Gear Ratio (-1,0,..7)
TranActGearRato (-1,0,..7)
Transmission Current Gear (-1,0,..7)
TransCurGear (-1,0,..7)
Transmission Driveline Engaged (0/1)
TransDrivEng (0/1)
Transmission Oil Temperature (F)
TransOilTemp (F)
Transmission Output Shaft Speed (Revolutions Per Minute (RPM))
TransOptShaftSpeedTrans_1587 (Revolutions Per Minute (RPM))
Transmission Torque Converter Lockup Engaged (0/1)
TransTorqConvLockupEngaged (0/1)
Transmission Range Attained (-1,0,..7)
TrRgAttai (-1,0,..7)
Transmission Selected Gear (-1,0,..7)
TrSelGr (-1,0,..7)
Vehicle Speed (miles/hour)
VehSpeedEng (miles/hour)
Density

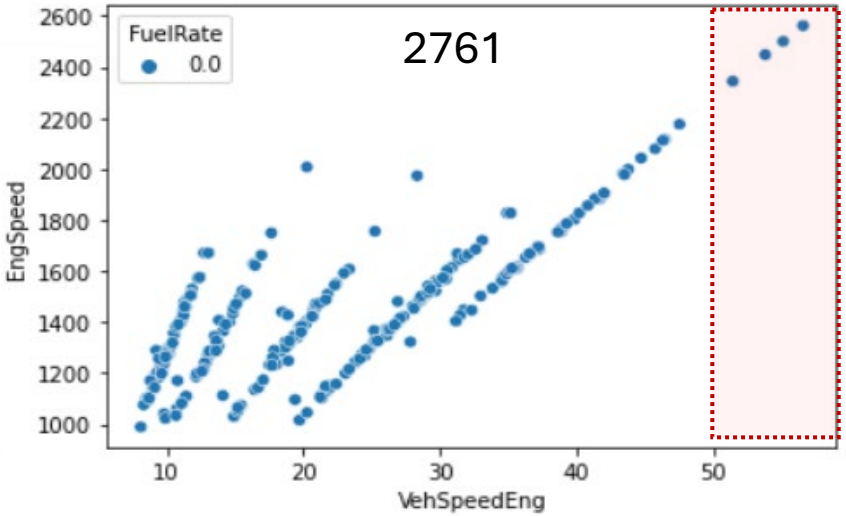
Engine Breaking Issue
EngRetarderStat_1587 ==1

EngRetarderStat_1587 ==1, it should slow down, but the 2744,2761 not slowing down. It is a safety related problem.









What variable actually carries the information?

Best Proxy is

Act_RetardPctTorqExh

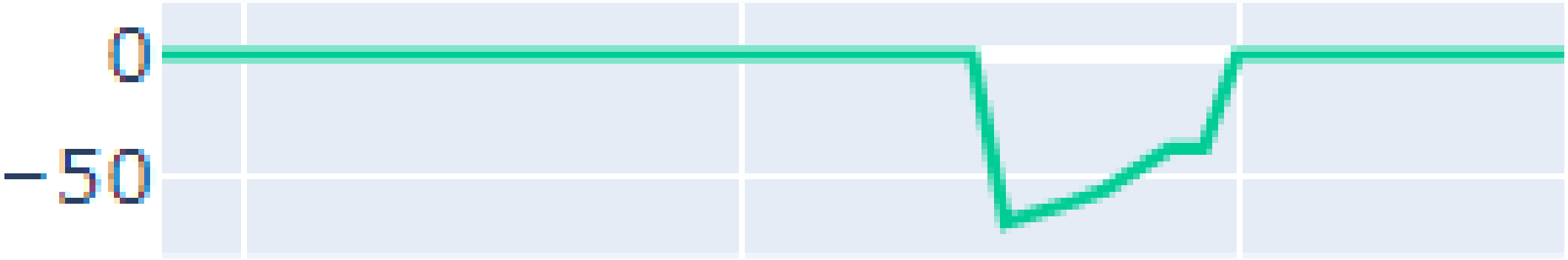
This is effectively:

How much negative torque the engine can generate at a given RPM

A healthy system will show **strong, predictable dependence on RPM**

A degraded system will **decouple** from RPM.

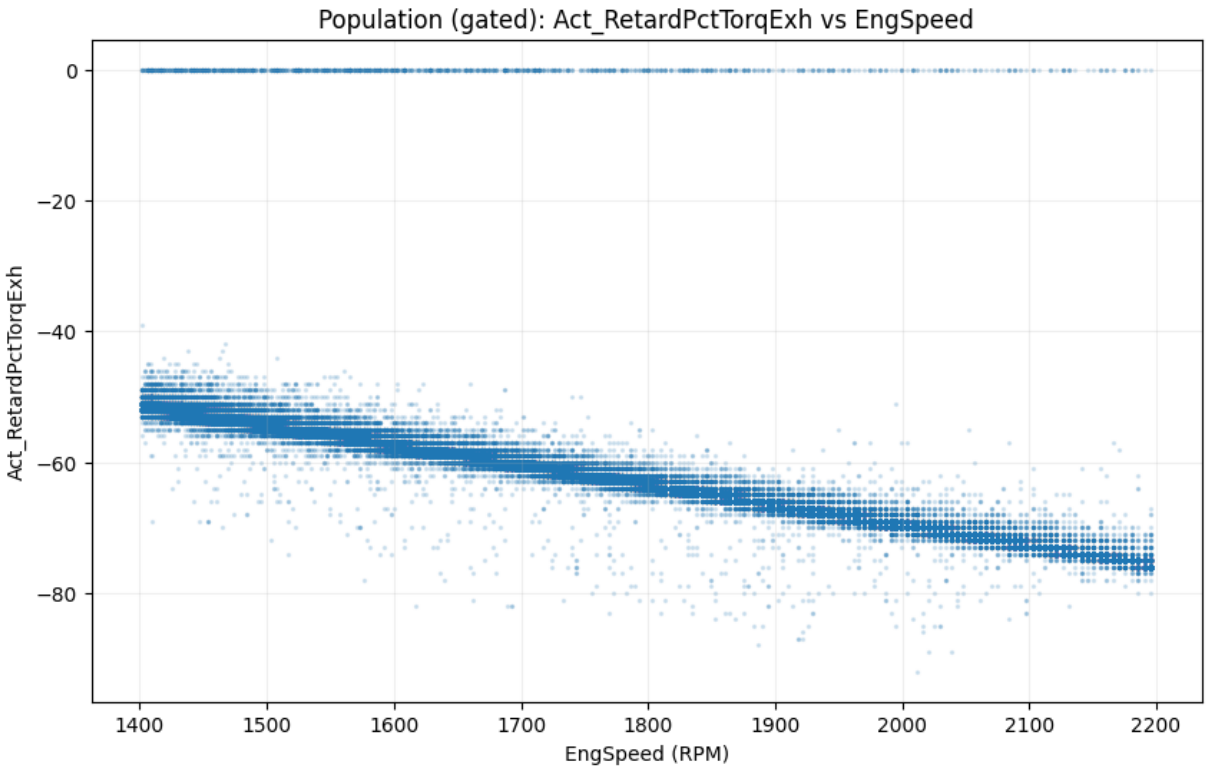
Key Channel Act_RetardPctTorqExh

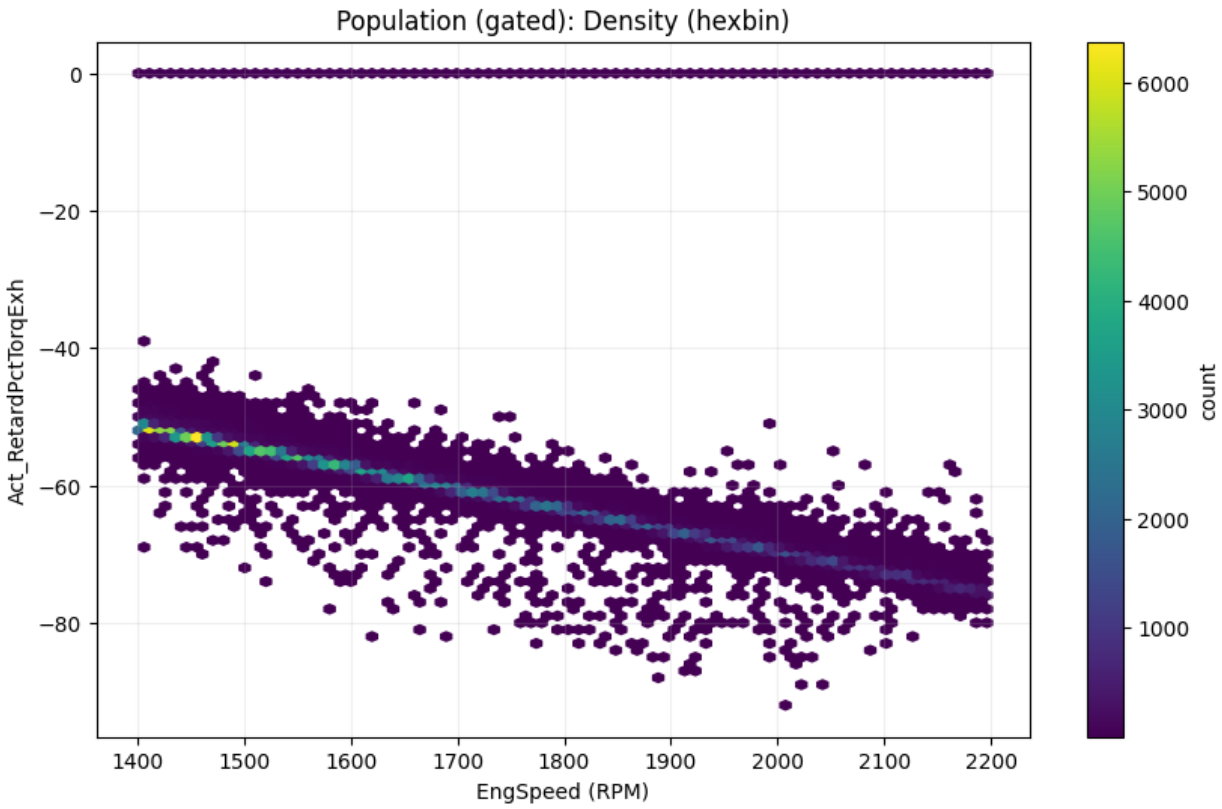


-100 % < Act_RetardPctTorqExh < 0%

The key signal is Act_RetardPctTorqExh

The unit is % here. -100% means, all 100% of the exhaust gas is going back into the engine due to closure of the valve. As soon as you close the valve, there will be a high pressure built up in the combustion chamber slowing down the vehicle. Due to lack of fuel for burring, this back pressure will go down to eventually zero.


Population (gated): Act_RetardPctTorqExh vs EngSpeed
Act_RetardPctTorqExh
EngSpeed (RPM)
0
−20
−40
−60
−80
1400
1500
1600
1700
1800
1900
2000
2100
2200


Population (gated): Density (hexbin)
Act_RetardPctTorqExh
EngSpeed (RPM)
count
0
−20
−40
−60
−80
1400
1500
1600
1700
1800
1900
2000
2100
2200
6000
5000
4000
3000
2000
1000

Population (gated): Pearson correlation heatmap
Act_RetardPctTorqExh
EngSpeed
VehSpeedEng
TrRgAttai
FuelRate
EngPctTorq
BoostPres
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
r

	Act_RetardPctTorqExh	EngSpeed	VehSpeedEng	TrRgAttai	FuelRate	EngPctTorq	BoostPres
Act_RetardPctTorqExh	1.000000	-0.823942	-0.522093	-0.053261	-0.001119	0.099181	0.006301
EngSpeed	-0.823942	1.000000	0.629779	0.065162	0.001079	-0.009361	0.001484
VehSpeedEng	-0.522093	0.629779	1.000000	0.756574	0.000556	-0.007835	-0.000396
TrRgAttai	-0.053261	0.065162	0.756574	1.000000	-0.000563	-0.004127	-0.001477
FuelRate	-0.001119	0.001079	0.000556	-0.000563	1.000000	-0.000098	0.021084
EngPctTorq	0.099181	-0.009361	-0.007835	-0.004127	-0.000098	1.000000	0.001364
BoostPres	0.006301	0.001484	-0.000396	-0.001477	0.021084	0.001364	1.000000

Data

UNRESTRICTED

Unzip Zip File

```
Project\data\raw\
├ Group1\
│   ├ *.cdf
│   └ ...
├ Group2\
│   ├ *.cdf
│   └ ...
```

- Group1
  - MODEL0018
  - MODEL0021
  - MODEL0029
  - MODEL0034
  - MODEL0042
  - MODEL0094
- Group2
  - MODEL0018
  - MODEL0021
  - MODEL0022
  - MODEL0028
  - MODEL0029
  - MODEL0034
  - MODEL0042
  - MODEL0095
  - MODEL0096

- Group1
  - MODEL0018
    - VIN02764
      - 2020
      - 2021
    - VIN02765
    - VIN02766
    - VIN02767
    - VIN02777

Name	Status	Date modified
MODEL0018_VIN02764_20201023_1357_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201105_1609_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201106_1133_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201107_1440_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201108_0025_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201109_1914_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201110_1826_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201111_1249_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201112_0054_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201113_0022_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201114_0008_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM
MODEL0018_VIN02764_20201115_1130_FAMILY0003_B_TS_Daily.cdf		10/11/2021 1:24 PM

Folder Structure

```
Project\
│
├─ myenv\                ← virtual environment (next step)
├─ data\
│  ├─ raw\              ← CDF files go here
│  └─ processed\        ← cleaned / derived data
├─ notebooks\
├─ src\
```

File Conversion

- from spacepy import pycdf

Cdf files

- --- Group1 --- ✅ Found 7343 CDF files First file: MODEL0018_VIN02764_20201023_1357_FAMILY0003_B_TS_Daily.cdf Sample path breakdown → group=Group1, model=MODEL0018, vin=VIN02764, year=2020 Variables (152 total): ['UTC_1Hz', 'BrakeSwitch', 'CrankcasePres', 'TrOutShaftSp', 'EngSpeed', 'FuelRate', 'IntManfTemp', 'PTOGovernorSetSwtchEng', 'ASROffRoadSwtchTire', 'PTOResSwtch_1587’]

- --- Group2 --- ✅ Found 13981 CDF files First file: MODEL0018_VIN02758_20160203_1612_FAMILY0003_B_TS_Daily.cdf Sample path breakdown → group=Group2, model=MODEL0018, vin=VIN02758, year=2016 Variables (149 total): ['UTC_1Hz', 'BrakeSwitch', 'CrankcasePres', 'TrOutShaftSp', 'PTOCtlSwt_1587', 'EngSpeed', 'FuelRate', 'EngOilPres', 'IntManfTemp', 'PTOGovernorSetSwtchEng'] (myenv) PS C:\Users\ggugaratshan\OneDrive - HBK\Documents\2_SIADS\SIADS699\Project>

Group1: **7,343** CDF files; Group2: **13,981** CDF files

Create a FAST “manifest” of all files (no CDF reading)

group, model, vin, year
date/time parsed from filename
full file path

This will scan filenames and paths and build a table with:

- group, model, vin, year
- date/time parsed from filename
- full file path

src/build_manifest.py → data/processed/cdf_manifest.csv

Extract Signals

- Each **CDF file** has:
  - a **header schema** = list of signal names (variables), Schema is **set of signal names** present in one CDF file
  - associated metadata (VIN, model, date)
- We need to answer **three questions**:
  - **Within a VIN**
    → Are signal names **consistent across time**?
  - **Across VINs (same model or group)**
    → Do different VINs have **different signal sets**?
  - **Over time**
    → Did signal names **appear/disappear/change** for a VIN (schema drift)?

- This is essentially **schema versioning & drift detection**, which is very common in this type of onboard diagnostic data.

High-level strategy on Signals

Phase 1 — Define the "schema unit" -set of signal names present in one CDF file

Phase 2 — Establish a baseline schema per VIN

- Collect all files for the vin
- Baseline schema, earliest file, or most common

Phase 3 — Compare every file against the baseline (within VIN)

- Missing signals: Signals present in the baseline but absent in this file
- New signals: Signals not in baseline but present in this file
- Unchanged signals: Intersection
- "Are signal names the same across all files for the same VIN?"

Phase 4 — Detect schema drift over time (within VIN) - Now add time awareness:

- For each VIN, Sort files by timestamp
- Track when:
  - a signal first appears
  - a signal disappears permanently
  - a signal flaps (appears, disappears, reappears)
- This produces: drift timelines, version boundaries, and change points
- Build a VIN × signal × time presence matrix.
  - This is where domain knowledge comes in later — but your pipeline should surface the facts cleanly.

Phase 5 — Compare schemas across VINs

- Once each VIN is internally consistent (or characterized): We compare VIN-level schemas, VIN A vs VIN B, common signals, VIN-specific signals
- This answers: "Are different vehicles instrumented differently?", We do this, within same MODEL, within same GROUP, across years

Phase 6 — Classify discrepancies (important for interpretation), Not all differences are "errors".

- We can classify signal differences as:
  - Expected differences
    - model-specific
    - optional equipment
    - year-based upgrades
  - Potential data quality issues
    - missing signals mid-life
    - unexpected renaming
    - inconsistent availability

Output from the Signal Work

- For each VIN: baseline signal list, % of files matching baseline, list of drift events with dates
- Across VINs: signals common to all, VIN-specific signals, signals that changed naming over time
- This is **good** for feature engineering, modeling assumptions, explaining results to stakeholders

1st Standup

Team 24: guga-fayrooz-seungdo-vishred

SIADS 699 - Team BrakeGuard – Truck Fleet Failure Prediction

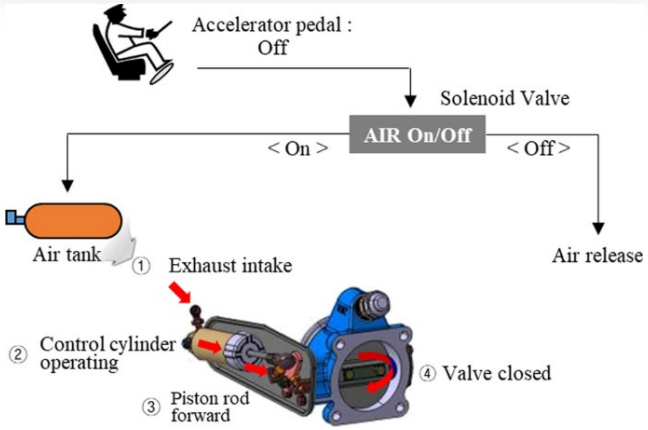
Project Description

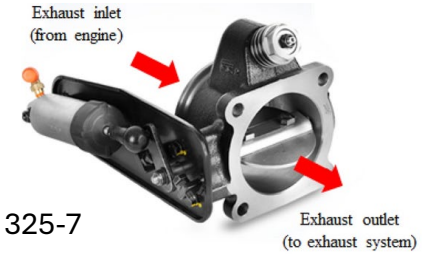
Trucks have two types of brake. One is the friction-type brake, similar to our car brakes. Another type of brake is These devices achieve braking force by blocking the exhaust pipe, thereby compressing the exhaust gas and consuming kinetic energy.

Failure of the exhaust brake in trucks are very risky and catastrophic and truck fleet operators want to know if the braking system is going to fail ahead of time to do preventive maintenance.

Early detection of exhaust brake system degradation using onboard sensor telemetry data (onboard sensor and diagnostic data) from a diesel truck fleet.

- Can we detect early signs of exhaust brake degradation before a field failure occurs?
- How do supervised and unsupervised learning methods compare in identifying anomalous braking behavior?
- Can we construct machine learning method or just machine learning methods to enable mask to isolate valid braking events and improve detection accuracy?
- How can cost-sensitive modeling better reflect real-world fleet operational risks





(a)

(b)

https://link.springer.com/article/10.1007/s42835-022-01325-7

What did the team do last week?

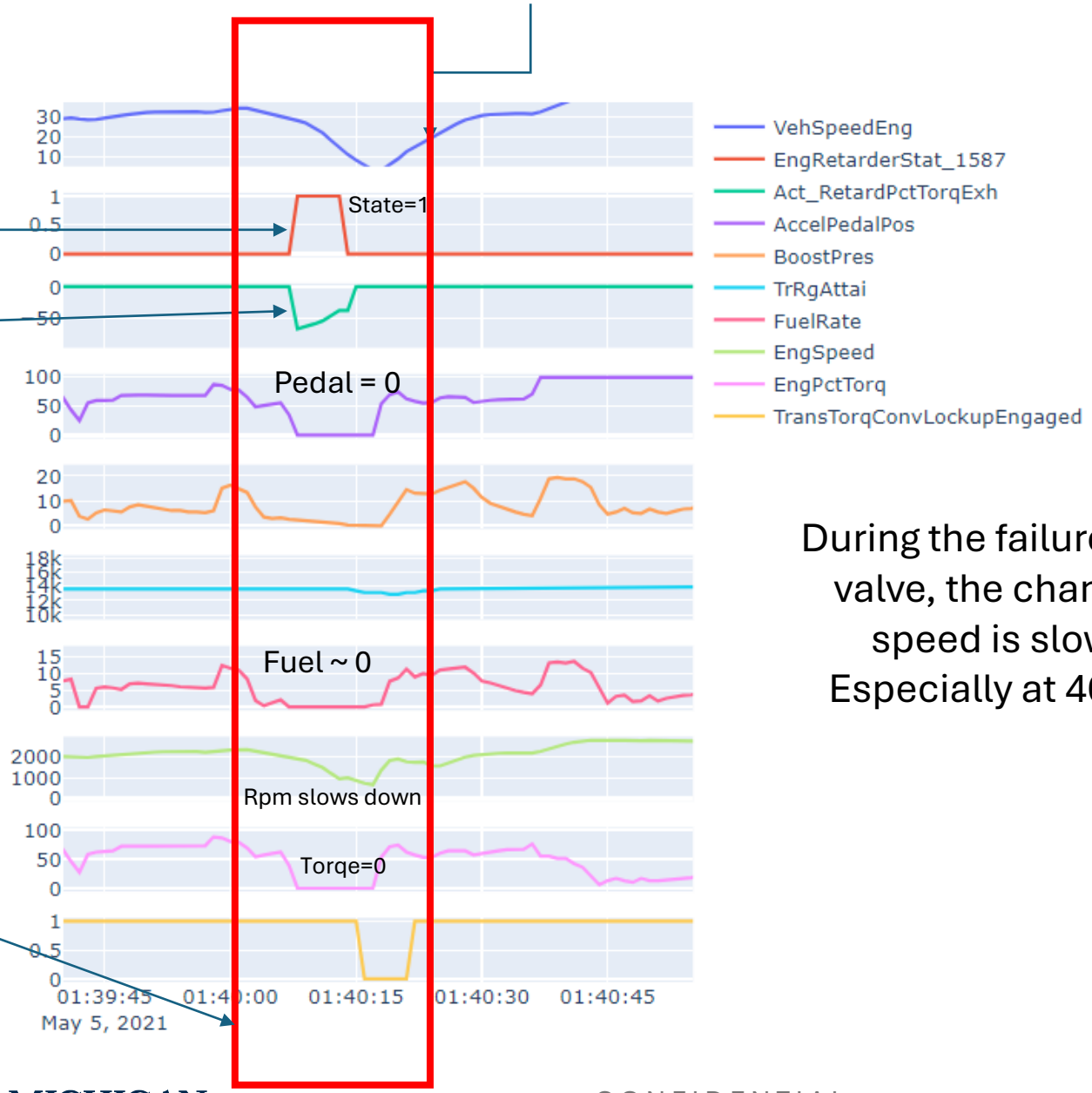
- Large amount of data
  - >90 vehicle data multiple years
- Worked on
  - Combining the data
  - Data cleaning
  - Data processing
  - Segmenting data for braking event

What are we working now?

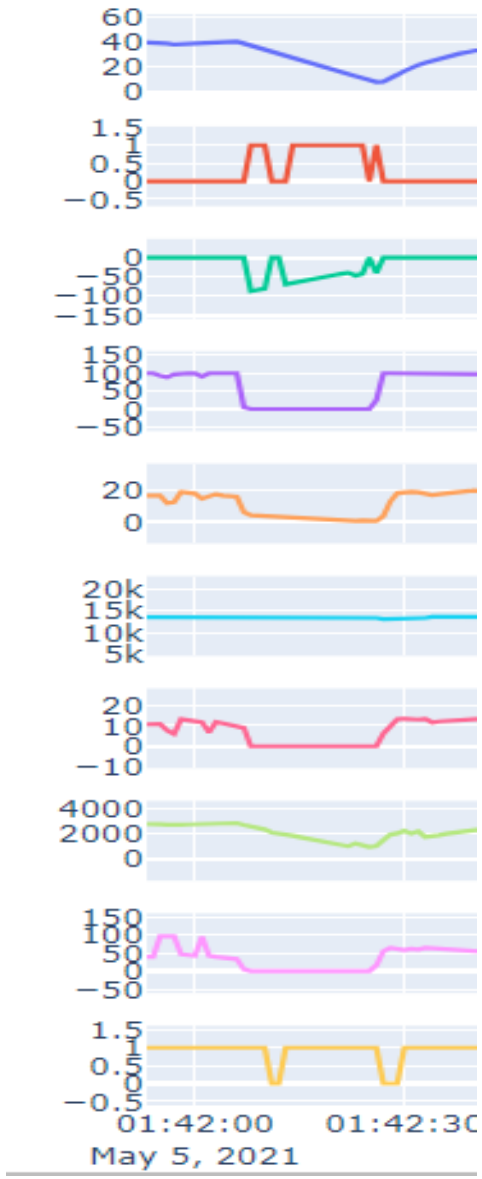
Signals

Engine Retarder is on

Exhaust back pressure Is budling up and then reduces due to zero fuel



During the failure of the valve, the change in speed is slower Especially at 40mph



Details in this window of data

What issues are blocking?

No Blockers

- Team is three different time zones?
  - US, Middle East, Asia
- Model training data and data leak
- Planning to discuss with the mentor